Sai Dhruv Yellanki

College Park, Maryland (Ready to Relocate)
240-960-1300 | dhruvsai@umd.edu | linkedin.com/in/saidhruvyellanki | github.com/dhruvyellanki19 | Portfolio

Professional Summary

AI Engineer and Data Science Graduate student with 3+ years of experience in fine-tuning LLMs (LoRA,QLoRA, RLHF), building Retrieval-Augmented Generation (RAG) systems, and developing multi-agent workflows for intelligent automation, predictive analytics, and anomaly detection. Proficient in Python, PySpark, Azure Databricks, and Kubernetes, with domain expertise across healthcare, finance, supply chain, and enterprise automation.

Education

University of Maryland — **August 2024 – May 2026**
Masters of Science in Data Science; GPA: 3.89/4.0 — *College Park, MD, USA*

Experience

Aurelius Tech & Talent Solutions — **May 2020 – August 2020**
AI Engineer Intern — *Ellicott City, MD, USA*

- Developed a proposal generation system integrating RAG pipeline with Azure OpenAI embeddings PostgreSQL pgvector and Redis caching for semantic search across 10K+ documents, reducing retrieval time by 65% and achieving 500 ms latency.
- Fine-tuned the GPT-OSS-20B model as an orchestrator in a Chain-of-Thought multi-agent framework, coordinating agents build using GPT-4 for retrieval, drafting, validation, and compliance, achieving 79% precision and 74% recall on compliance-detection.
- Deployed the system through containerized LLMOps workflows using Docker, Kubernetes, and Azure DevOps CI/CD, automating deployment, scaling, and monitoring to maintain 95% uptime and shorten release cycles by 40%.
- Optimized system performance with distributed processing and Redis caching, increasing document throughput by 35% and enabling real-time integration of structured and unstructured data to improve proposal-generation speed and accuracy.

Watzmann Consulting — **June 2023 – June 2024**
Data Analyst – Predictive Analytics & Machine Learning — *Hyderabad, TG, India*

- Engineered GDPR-compliant data pipelines in Azure Databricks using PySpark and SQL to unify and standardize customer, system, and operational data, improving reliability and enabling scalable analytics across 200K+ records.
- Designed a demand forecasting solution using XGBoost, LightGBM, and Bi-LSTM on historical and behavioral data to predict sales and capacity needs 12% more accurately (Mean Absolute Error less than 0.6), enabling proactive inventory and staffing decisions.
- Created a workload optimization system that forecasted job runtimes with Bi-LSTM and applied mixed-integer linear programming optimization to schedule resources efficiently, reducing idle compute time by 18% and cutting infrastructure costs.
- Implemented MLOps pipelines with MLflow and Azure Machine Learning to automate model retraining and deployment, and deliver Power BI dashboards for stakeholder monitoring, boosting model reliability and operational efficiency by 15%.

Ramphal Technologies — **December 2022 – May 2023**
Data Science Intern — *Hyderabad, TG, India*

- Trained system-performance prediction models using PyTorch and XGBoost to forecast application latency and server load, improving prediction accuracy by 12% and reducing average request delay by 7.5 ms.
- Constructed a real-time event pipeline with RabbitMQ and PostgreSQL to process 10K+ telemetry events per second, reducing false alerts and cutting reporting latency by 30% through Tableau dashboards and improved operational visibility for engineering teams.

Projects

LoRA-Tuned RLHF News Summarization System | *RLHF, RAG, LORA, Docker, Kubernetes, Grafana, vLLM* | GitHub Link

- Built a news-summarization system using RAG with PostgreSQL pgvector for retrieval and a T5 model fine-tuned using human feedback reinforcement learning (trlx, PPO) on CNN News data, improving ROUGE-L by 11% and reducing hallucinations by 6%.
- Orchestrated model deployment as a scalable API with Docker, FastAPI, vLLM, and Kubernetes, integrating CI/CD pipelines and a Netlify web app to deliver accurate summaries with 240 ms p95 latency.

Multi-Agent Code Automation System | *Python, Mistral-7B-Instruct, LangGraph, vLLM, LLM Agent Orchestration* | GitHub Link

- Engineered a multi-agent code-generation system using Mistral-7B-Instruct and LangGraph, where specialized agents autonomously planned, coded, and debugged applications, automating 85% of development tasks and cutting delivery time by 4x.
- Executed local deployment using vLLM for optimized inference, a controlled sandboxed environment for safe code execution, and a Streamlit interface for user interaction, achieving an 83 % generation success rate and 90 % faster development cycles.

Technical Skills

Programming Languages & Databases: Python (Pandas, NumPy, Scikit-learn, PySpark, TensorFlow, PyTorch, SpaCy, NLTK, Matplotlib, Seaborn, Plotly, Dash), SQL, R,PostgreSQL, Redis, MySQL, MongoDB.

Tools & Technologies: Docker, Kubernetes, Azure (Machine Learning, DevOps, Databricks),AWS(Bedrock, Sagemaker) MLflow, FastAPI, vLLM,Git, Power BI, Tableau, Grafana, CI/CD Pipelines.

Machine Learning & Analytics: Predictive Modeling, Deep Learning, Time Series Forecasting (Prophet, ARIMA), MLOps, model stacking, Model Deployment & Monitoring, Anomaly Detection, hyperparameter tuning, cross-validation.

AI & LLM Engineering: NLP (NER, summarization, sentiment analysis, embeddings, transformers), fine-tuning(LoRA,QLoRA, RLHF), RAG, multi-agent orchestration (LangChain, LlamaIndex, LangGraph, AutoGen), Prompt Engineering, inference (vLLM).